

Topic 8: Redox Chemistry and Groups 1, 2 and 7

8A: Redox chemistry

Students will be assessed on their ability to:

8.1	know what is meant by the term 'oxidation number' and understand the rules for assigning oxidation numbers
8.2	be able to calculate the oxidation number of elements in compounds and ions, including in peroxides and metal hydrides
8.3	be able to indicate the oxidation number of an element in a compound or an ion, using a Roman numeral
8.4	be able to write formulae given oxidation numbers
8.5	understand oxidation and reduction in terms of electron transfer and changes in oxidation number, and the application of these ideas to reactions of <i>s</i> -block and <i>p</i> -block elements
8.6	know that oxidising agents gain electrons and reducing agents lose electrons
8.7	understand that a disproportionation reaction involves an element in a single species being simultaneously oxidised and reduced
8.8	know that oxidation number is a useful concept in terms of the classification of reactions as redox and as disproportionation
8.9	understand that metals, in general, form positive ions by loss of electrons with an increase in oxidation number whereas non-metals, in general, form negative ions by gain of electrons with a decrease in oxidation number
8.10	be able to write ionic half-equations and use them to construct full ionic equations

8B: The elements of Groups 1 and 2

Students will be assessed on their ability to:

8.11	understand reasons for the trend in ionisation energy down Groups 1 and 2
8.12	understand reasons for the trend in reactivity of the elements down Group 1 (Li to K) and Group 2 (Mg to Ba)
8.13	know the reactions of the elements of Group 1 (Li to K) and Group 2 (Mg to Ba) with oxygen, chlorine and water
8.14	know the reactions of: i oxides of Group 1 and 2 elements with water and dilute acid ii hydroxides of Group 1 and 2 elements with dilute acid
8.15	know the trends in solubility of the hydroxides and sulfates of Group 2 elements
8.16	understand the reasons for the trends in thermal stability of the nitrates and the carbonates of the elements in Groups 1 and 2 in terms of the size and charge of the cations involved

8.17	understand the formation of characteristic flame colours by Group 1 and 2 compounds in terms of electron transitions <i>Students will be expected to know the flame colours for Group 1 and 2 compounds.</i>
8.18	know experimental procedures to show: i patterns in the thermal decomposition of Group 1 and 2 nitrates and carbonates <i>Students will be expected to know tests for carbon dioxide and oxygen; and to recognise nitrogen dioxide by its colour and acidic pH.</i> ii flame colours in compounds of Group 1 and 2 elements
8.19	know reactions, including ionic equations where appropriate, for identifying: i carbonate ions, CO_3^{2-} , and hydrogencarbonate ions, HCO_3^- , using an aqueous acid to form carbon dioxide (and testing the gas with limewater) ii sulfate ions, SO_4^{2-} , using acidified barium chloride solution iii ammonium ions, NH_4^+ , using sodium hydroxide solution and warming to form ammonia (and testing with litmus and HCl fumes)
8.20	be able to calculate solution concentrations, in mol dm^{-3} and g dm^{-3} , including simple acid-base titrations using the indicators methyl orange and phenolphthalein
8.21	CORE PRACTICAL 3 Finding the concentration of a solution of hydrochloric acid.
8.22	understand how to minimise the sources of measurement uncertainty in volumetric analysis and estimate the overall uncertainty in the calculated result
8.23	CORE PRACTICAL 4 Preparation of a standard solution from a solid acid and use it to find the concentration of a solution of sodium hydroxide.
	Further suggested practicals: i experiments to study the thermal decomposition of Group 1 and 2 nitrates and carbonates ii flame tests on compounds of Group 1 and 2 iii simple acid-base titrations using the indicators methyl orange and phenolphthalein to calculate solution concentrations in g dm^{-3} and mol dm^{-3} iv the solubility of calcium hydroxide by titration v determination of moles of water of crystallisation by titration

8C: Inorganic chemistry of Group 7 (limited to chlorine, bromine and iodine)

Students will be assessed on their ability to:

8.24	understand reasons for the trends for Group 7 elements in: i melting and boiling temperatures and physical state at room temperature ii electronegativity iii reactivity down the group
8.25	understand the trend in reactivity of Group 7 elements in terms of the redox reactions of Cl_2 , Br_2 and I_2 with halide ions in aqueous solution <i>Students are expected to know the colours of the elements in standard conditions, in aqueous solution and in a non-polar organic solvent.</i>
8.26	understand, in terms of changes in oxidation number, the following reactions of the halogens: i oxidation reactions with Group 1 and 2 metals ii the disproportionation reaction of chlorine with water and the use of chlorine in water treatment iii the disproportionation reaction of chlorine with cold, dilute aqueous sodium hydroxide to form bleach iv the disproportionation reaction of chlorine with hot alkali v reactions analogous to those specified above
8.27	understand the following reactions: i solid Group 1 halides with concentrated sulfuric acid, to illustrate the trend in reducing ability of the hydrogen halides ii precipitation reactions of the aqueous anions Cl^- , Br^- and I^- with aqueous silver nitrate solution and nitric acid, and the solubility of the precipitates in aqueous ammonia solution iii hydrogen halides with ammonia gas (to produce ammonium halides) and with water (to produce acids)
8.28	be able to make predictions about fluorine and astatine and their compounds, in terms of knowledge of trends in halogen chemistry
	Further suggested practicals: i reaction of solid potassium halides with concentrated sulfuric acid ii precipitation reaction for halides and other anions